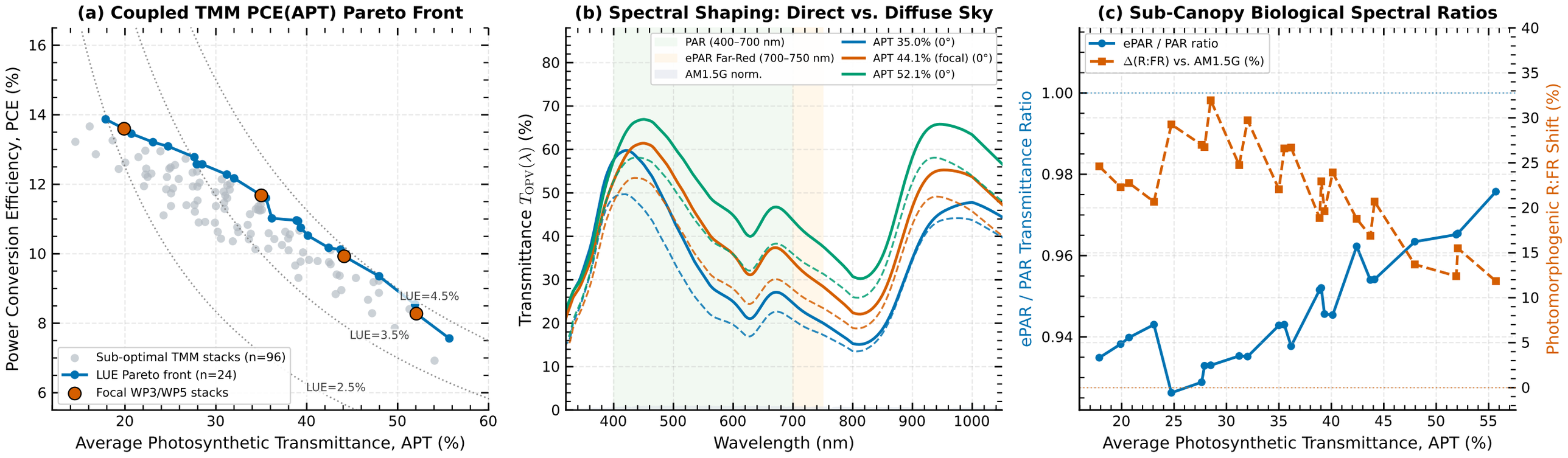

(a) Coupled TMM PCE(APT) Pareto Front
Power Conversion Efficiency, PCE (%)
Average Photosynthetic Transmittance, APT (%)
Sub-optimal TMM stacks (n=96)
LUE Pareto front (n=24)
Focal WP3/WP5 stacks
LUE=4.5%
LUE=3.5%
LUE=2.5%
(b) Spectral Shaping: Direct vs. Diffuse Sky
Transmittance $T_{\mathrm{OPV}}(\lambda)$ (%)
Wavelength (nm)
PAR (400–700 nm)
ePAR Far-Red (700–750 nm)
AM1.5G norm.
APT 35.0% (0°)
APT 44.1% (focal) (0°)
APT 52.1% (0°)
(c) Sub-Canopy Biological Spectral Ratios
ePAR / PAR Transmittance Ratio
Photomorphogenic R:FR Shift (%)
Average Photosynthetic Transmittance, APT (%)
ePAR / PAR ratio
Δ(R:FR) vs. AM1.5G (%)